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Module 4 Glossary: Unsupervised Learning and Generative Models in Keras

Welcome! This alphabetized glossary contains many terms used in this course. Understanding these terms is essential when working in the industry, participating in user groups, and participating in other certificate programs.

Term	Definition
Anomaly detection	A type of unsupervised learning used to identify unusual data points that do not fit the general pattern in a dataset.
Adam Optimizer	An optimization algorithm that can be used instead of the classical stochastic gradient descent procedure to update network weights iteratively based on training data.
Adversarial training	A training method in GANs where the generator and discriminator networks are trained simultaneously, with the generator aiming to fool the discriminator, and the discriminator trying to accurately classify real and fake data.
Autoencoders	A type of neural network used to learn efficient representations of data, often for dimensionality reduction or feature learning.
Binary crossentropy loss	A loss function used in binary classification tasks, often utilized in training neural networks to measure the difference between predicted and actual outputs.
Bottleneck	The central, most compressed layer in an autoencoder that contains the most critical features of the input data.
Clustering	A method in unsupervised learning that involves grouping data points into clusters, where data points in the same cluster are more similar to each other.
Convolutional autoencoders	A type of autoencoder that uses convolutional layers, making it particularly effective for tasks involving image data.
Convolutional neural network (CNN)	A type of deep neural network commonly used in image processing tasks, known for its ability to capture spatial hierarchies in images.
Data augmentation	A technique used to increase the diversity of data available for training models by generating new synthetic data, often by applying transformations to existing data.
Decoder	The part of an autoencoder that reconstructs the input data from the compressed latent space representation.
Denoising	The process of removing noise from data, such as images, to improve their quality.
Diffusion	A physical process where particles spread from regions of high concentration to low concentration; in diffusion models, this concept is simulated to generate or enhance data.
Diffusion model	A type of probabilistic generative model that iteratively refines noisy data to produce high-quality samples, often used in image generation.
Dimensionality reduction	A process in unsupervised learning that reduces the number of random variables under consideration by obtaining a set of principal variables.
Discriminator network	In a generative adversarial network (GAN), this network evaluates the authenticity of the generated data, distinguishing between real and fake data.
Encoder	The part of an autoencoder that compresses the input data into a latent-space representation.
Epochs	In machine learning, an epoch refers to one complete pass of the training dataset through the learning algorithm.